SRILAKSHMI S

Software Engineer | Python • SQL • Flask • REST APIs • Debugging

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate (B.E., CGPA 8.0) with strong Python and SQL skills and hands-on experience building, testing, debugging, and deploying software. Built and deployed three end-to-end Python projects using Flask, XGBoost, CNN, and OpenCV, including a bug-analysis platform that applies static code analysis to detect and explain defects. Completed a project-based internship working with 10,000+ records, SQL queries, and Python-based data processing. Seeking a Software Engineer role to write clean, maintainable code and contribute to reliable software in an Agile team.

TECHNICAL SKILLS

Programming: Python (primary), SQL

Web & APIs: Flask, REST APIs, web application deployment (Render)

Databases & Data: SQL (querying, filtering, aggregation, joins), Pandas, NumPy

Tools & Version Control: Git, GitHub, Roboflow, Power BI, Microsoft Excel

ML & Libraries: XGBoost, CNN, OpenCV, Machine Learning

Cloud: AWS (Cloud Architecting)

Concepts: Debugging, static code analysis, data cleaning, EDA, SDLC

PROJECTS

AI-Assisted Python Code Quality and Bug Analysis Platform | https://new-sri.onrender.com *Python, Flask, XGBoost, Static Code Analysis*

- Developed and deployed a Flask web application that analyzes user-submitted Python code and returns predicted bug types, severity levels, explanations, and suggested fixes.

- Extracted code metrics using static analysis (lines of code, cyclomatic complexity, maintainability index) and built an XGBoost model achieving 92% accuracy and 92% F1-score across 100 bug categories.
- Collected and preprocessed 20,000 labelled Python code samples to train and validate the bug prediction model.

Detection of Diabetic Retinopathy *Python, CNN, Flask*

- Designed and trained a CNN to classify diabetic retinopathy into five severity stages from fundus images, achieving 96% accuracy.
- Applied image preprocessing, normalization, and data augmentation, and tuned hyperparameters to reduce misclassification.
- Deployed the model in a Flask web application for real-time prediction.

Enhanced Haar Cascades for Face Detection *Python, OpenCV*

- Implemented an OpenCV Haar Cascade face detector on annotated grayscale datasets, achieving 90% accuracy.
- Fine-tuned detection parameters (scale factor, minimum neighbors, size limits) to reduce false positives while keeping the solution efficient.
- Evaluated results across varying lighting conditions and scales, showing improved accuracy over default settings.

PROFESSIONAL EXPERIENCE

Data Analytics Intern | Doosan Bobcat Pvt Ltd, Chennai | Jun 2025 – Sep 2025 | https://github.com/Sri0295/Sales-Data-Analysis

- Wrote Python (Pandas, NumPy) scripts to clean and validate 10,000+ sales records, removing duplicates and handling missing values to improve data quality.
- Developed SQL queries across sales, products, complaints, and tickets tables for faster data retrieval and KPI analysis.
- Performed EDA and statistical analysis (outliers, skewness, correlation), uncovering a strong Sales–Profit relationship (r = 0.83).
- Built Excel reports and a 3-page Power BI dashboard with interactive KPIs, identifying Laptop (22% of sales) and West region (28% of sales) as key contributors.

EDUCATION

B.E. Computer Science Engineering | PMC Tech, Hosur | CGPA: 8.0 | 2022 – 2026

CERTIFICATIONS

- AWS Academy Graduate – Cloud Architecting
- CS50P: Introduction to Programming with Python – Harvard University
- IBM Digital Badge in Data Science – IBM SkillsBuild / Credly
- Data Visualization: Empowering Business with Effective Insights – Tata (Forage Job Simulation)